

MUSCLE: Multi-Attribute User Session for Query Classification

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Abstract

Disambiguating broad user intent is an important challenge in e-commerce, as multiple possible meanings of ambiguous queries can result in completely different product selections. Such queries can be disambiguated in the context of the *shopping session*, as prior interactions provide crucial cues about users’ underlying preferences. Existing query classification models ignore the cues from shopping sessions, effectively reducing the broad scope of possible customer preferences to a single interpretation.

To address this limitation, we propose **MUSCLE (Multi-attribute User Session for query Classification)**— a query classification framework leveraging the shopping session context to disambiguate broad intent and capture evolving user preferences. Our approach views user sessions as diverse sequences of queries, item interactions and contextual features, integrating semantic signals from queries with session-level behavioral patterns.

We demonstrate that MUSCLE consistently outperforms state-of-the-art models, which ignore either the historical shopping behavior or the current search context. We show that by capturing both immediate query semantics and broader session context, MUSCLE consistently improves query classification on brand and product type prediction tasks, while remaining a practical solution for real-life applications.

Keywords

E-commerce query classification, shopping sessions, brand recommendation

1 Introduction

E-commerce search queries are often short and ambiguous, allowing for multiple interpretations of a true user intent. This ambiguity poses a significant challenge for e-commerce stores, particularly in search **query classification** task, whose goal is to accurately map a user’s query to the intended target class.

Previous **query understanding** approaches [10, 12, 14] attempt to infer user intent solely from the textual content of the short, potentially ambiguous queries. While these models excel at predicting the most popular or generic meaning of such queries, they fail to tailor their predictions for the underlying preferences of each individual user. For instance (Figure 1), a user searching for “yoga”

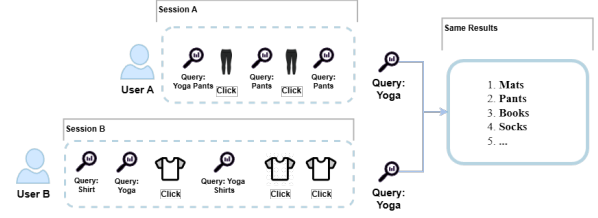


Figure 1: Two users with distinct session histories issue the same query, “Yoga.” Despite different histories, the system returns identical results, ignoring individual preferences.

may be interested in yoga pants or yoga t-shirts, and their preference can be captured by the previous shopping behavior, which the conventional query classification systems ignore.

Incorporating users’ previous shopping behavior lies within **next-item recommendation** paradigm and can be broadly categorized into sequence-based models that rely solely on interaction histories [1, 4, 5, 9, 13, 18], and side-information-aware models that incorporate item attributes, user metadata, or multimodal content, to improve recommendation quality [2, 6–8, 11, 15–17]. Yet, the purpose of these session-based models is to capture the evolution of historical interests, and they cannot be applied in cases when the user provides a situational context (search query).

As a result, the recommendations remain anchored to prior item events rather than the user’s current query intent. Consider a scenario in which a user issues a query “*running shoes*” and subsequently clicks on two product pages. Session models treat these clicked items as the dominant behavioral signal and generate predictions for the next item, typically recommending additional shoe models or related accessories. However, when the user later submits a new query, such as “*ankle support brace*”, these models lack a mechanism to reinterpret or integrate the updated query semantics. In contrast, we find that incorporating previous shopping actions can help distinguish varying intents for ambiguous queries.

We propose **MUSCLE (Multi-Attribute User Session for Query Classification)**, an e-commerce query classification model that leverages the user’s shopping session to disambiguate intent and capture evolving preferences. MUSCLE models user sessions as heterogeneous sequences combining **queries, item events, and contextual signals**, which allows to integrate semantic cues from the query with behavioral patterns from the session. The contributions of this work are as follows:

*This work was conducted during an internship at Amazon.

- (1) We propose *MUSCLE*, a search query classification method enhanced by the shopping session context;
- (2) We study the effects of incorporating heterogeneous shopping session information, which enables more accurate disambiguation of the user intent;
- (3) We conduct extensive experiments on anonymized session data to evaluate *MUSCLE* on **brand** and **product type** prediction tasks, demonstrating its effectiveness in improving query classification and disambiguation compared to the state-of-the-art methods.

2 Problem Statement and Formulation

We focus on the search query classification task for brand and product type intents. For simplicity, we introduce the problem formulation and model description with respect to the product type prediction task; for brand recommendation, the setting is similar.

We formulate the task as follows: given an input query q , predict the most likely product type (PT) of the item that the customer intends to purchase. To enhance the performance of conventional query classifiers, we condition the product type prediction on the user’s shopping session. We model the session as a sequence of heterogeneous events:

$$S = (e_1, e_2, \dots, e_T), \quad (1)$$

where each event $e_t \in \mathcal{E}$ is a tuple:

$$e_t = (a_t, q_t, p_t), \quad (2)$$

where $a_t \in \{\text{search}, \text{click}, \text{add-to-cart}, \text{purchase}\}$ is the user action, q_t is the query corresponding to the event at time step t , and p_t is the product type of the item. For $a_t = \text{search}$, the item is undefined and hence $p_t = \emptyset$. For all other actions, p_t corresponds to the product type of the clicked, added, or purchased item, and q_t is the last searched query in the session.

The objective is to predict the product type p_T of the last search query of the session q_T , given all previous session events, formally $P(p_T \mid S_{<T}, q_T)$. For conventional query classification task $T = 1$ (no session is used).

3 Methodology

We introduce *MUSCLE*, a session-aware query classification model that predicts a class associated with the user’s current query by integrating semantic information from the query with behavioral signals from the session (historical events). *MUSCLE* adopts a bidirectional Transformer architecture, trained with a masked-language-model (MLM) objective [3, 13], for modeling session context and enabling intent classification within heterogeneous user sessions.

3.1 Input Representation

MUSCLE operates over a sequence of event embeddings $emb(e_t)$:

$$emb_{e_t} = emb(a_t) \oplus emb(q_t) \oplus emb(p_t) \oplus emb(t), \quad (3)$$

where the symbol $\oplus$ represents an element-wise summation of embeddings. The operator $emb(\cdot)$ denotes the embedding functions corresponding to the input components listed below:

- **Action embedding** ($emb(a_t)$): a learned categorical embedding representing the type of the event (search, click, add-to-cart, purchase).

- **Query embedding** ($emb(q_t)$): an embedding of the user query’s text, obtained from a pretrained model.
- **Product type embedding** ($emb(p_t)$): a learned categorical embedding of the product type (or brand) associated with the interacted item. Notably, the embedded item product types align with the target labels used in the prediction task; that is, during training, the model learns to predict these same categories as it outputs at inference.
- **Positional embedding** ($emb(t)$): a learned embedding encoding the position t of the event in the session.

Alternative Text Embedding Variants. In addition to the basic query embedding, we explore two alternative variants that aim to enrich the textual representation of events:

- (1) **MUSCLE_{title}** – to form $emb(q)$, we concatenate the query with the title of the item that the customer interacted with (when available).
- (2) **MUSCLE_{PT}** – $emb(q)$ is the combination of the query with semantically similar product categories. We precompute such categories for each query by building an index over query embeddings from a held-out sample collected on a different day. For a given query, we retrieve semantically similar queries from the index whose similarity exceeds a threshold τ , and select the top- k queries along with their associated PTs. For example, for a query “yoga”, similar queries such as “yoga shoes” or “yoga attire” provide their PTs. The original query and retrieved PTs are embedded separately and summed: $emb_{e_t} \oplus emb(\text{PTs})$, enabling the representation to incorporate richer semantic context and related query signals. Importantly, as the similar queries come from a temporally disjoint sample, no target labels are leaked from the product type prediction task.

3.2 Model Architecture

We linearly project the components of emb_{e_t} to match a common dimensionality and sum them. Unlike prior work [19], we adopt a Transformer-based architecture and enable the integration of additional input components, allowing richer and more flexible modeling. The resulting sequence of event embeddings is processed by the Transformer encoder (Figure 2), producing contextualized representations h_t for each historical event and a contextual representation h_T for the current query q_T : $(emb_{e_1}, emb_{e_2}, \dots, emb_{e_t}) \mapsto (h_1, \dots, h_T)$. The model produces a probability distribution over the product type space. During training, we maximize the log-likelihood of the observed product type label. At inference, we select the predicted product type(s) as:

$$\hat{p}_T = \arg \max_p P(p \mid S_{<T}, q_T) \quad (4)$$

3.3 Training and Inference Configuration

3.3.1 Session-Based Data Splitting. Instead of a traditional hold-one-out split (where the last event of each user is held out for testing) [7, 13], we adopt a session-based splitting strategy. In our dataset, the training signal is organized by complete historical sessions, so we mirror that in our evaluation: entire sessions are assigned to training, validation, or test sets.

3.3.2 Training Procedure. MUSCLE employs a *masked modeling* training strategy [3, 13], adapted for session-aware query classification. For each session, we randomly mask the product type of some user actions with probability $p = 0.15$ and replace it with a special token [MASK]. The query text and other inputs, not directly tied to items, remain **unmasked**. This strategy reflects the inference scenario, in which the model observes the session context and the current query, and predicts the associated product type. During training, the model learns to predict the masked product type, given the remaining unmasked session context.

3.3.3 Inference. To predict the product type associated with the next user event, we append a [MASK] token to the end of the session sequence. Given the preceding session history $S_{<T}$ MUSCLE produces contextualized representations from the augmented sequence. The hidden state corresponding to the [MASK] token is then used to compute a probability distribution over the product type space: $P(p_T \mid S_{<T}, q_T)$. This procedure enables the model to generate **context-aware product type predictions** conditioned on the observed session and the current query.

4 Experiments

We present the results of a qualitative and quantitative assessment of our method through the following research questions:

- (1) **RQ1 Model Effectiveness:** Does the proposed model provide measurable improvements compared to the baselines?
- (2) **RQ2 Effect of Additional Features:** Does the inclusion of supplementary features lead to further performance gains?
- (3) **RQ3 Effect of Session Length and Computation Overhead:** To what extent does session length impact the performance, and what is the computation overhead of our model?

4.0.1 Experimental Setup. We utilize an experimental dataset of 300k anonymized sessions sampled from an e-commerce marketplace. The shopping sessions are constructed by identifying queries followed by add-to-cart events for the same item, retaining only sessions where the item is purchased, and retrieving all actions within a one-hour window preceding the purchase. We fix the session length to **10** events—the average within a one-hour window—to balance computation amount and accuracy.

We evaluate our model on **three tasks**: predicting *product type* (including hundreds of PTs), *brand* (including tens of thousands

of brands), and *top-10k brands* (10k brands by frequency in the *brand* dataset). We report **Recall@k**, **MRR@k**, and **NDCG@k** and compare our model against the **baselines** from previous studies:

- **Query-Only model [14]:** a supervised model that predicts the final purchase using only the current query’s semantics, ignoring session context.
- **Session-Only model** (following BERT4Rec [13]): session-based model that predicts purchases, capturing temporal patterns but not the current query’s semantic intent.

Both baselines are trained on the same data as MUSCLE; Session-Only uses the same architecture and training. All models use a four-layer, four-head fine-tuned Transformer encoder and a two-layer feed-forward classifier.

4.1 Experimental Results & Analysis

We report the evaluation results for baselines and all MUSCLE variants in Table 1.

4.1.1 Comparative Evaluation Against Baseline Models (RQ1). MUSCLE consistently outperforms the baseline models across all datasets and values of k , showing the benefit of jointly modeling query and session history. The Session-Only model is the weakest baseline on all prediction tasks, which underlines the importance of the situational context of the last query. On the *All-Brands* dataset, the overall performance is lower, as brand recommendation is a more ambiguous task than product type prediction, with several equivalent brands matching the customer intent, but MUSCLE still consistently leads across all metrics. Finally, we note that the metrics based on top-1 ranking (MRR, NDCG) saturate quickly, and increasing k has little effect on performance, because cross-entropy heavily favors putting the correct label at rank 1.

4.1.2 Impact of Additional Query Context (RQ2). We observe that both $MUSCLE_{title}$ and $MUSCLE_{PT}$ show performance gains compared to the vanilla version. The gains of expanded versions are more pronounced on the *All-Brands* dataset, because incorporating textual cues from item titles or product types can help make the ambiguous brand recommendation task more finely tuned to the user’s historical preferences.

4.1.3 Impact of Session Length (RQ3). The following experiments are performed using the basic version of MUSCLE. We analyze how *session length* affects prediction accuracy (Figure 3). When predicting product types, longer sessions do not consistently improve performance and can even be detrimental. This is not merely due to a larger label space, but because additional events often contribute little new discriminative information once a few informative interactions have occurred. Often, earlier events introduce noise as user behavior drifts or includes unrelated browsing. Moreover, product-type cues are often explicitly stated in the query itself, thereby limiting the marginal value of older session context.

In contrast, brand prediction benefits from longer sessions, as brand intent is often expressed through subtle behavioral patterns that accumulate over multiple interactions, such as repeated engagement or transitions among related brands. From a systems perspective, intermediate session lengths strike a practical balance by capturing meaningful behavioral signals while avoiding the high computational cost and noise associated with very long sessions.

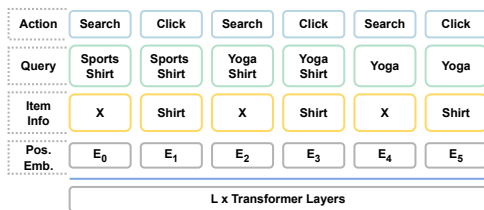


Figure 2: Overview of MUSCLE, a transformer encoder processes a sequence of user events to predict the next product type from session context and the current query.

Table 1: Performance comparison of MUSCLE variants and baselines across datasets. Best results per row shown in bold.

Dataset	k	MUSCLE			MUSCLE _{title}			Muscle _{PT}			Query-Only			Session-Only		
		R@k	N@k	M@k	R@k	N@k	M@k	R@k	N@k	M@k	R@k	N@k	M@k	R@k	N@k	M@k
Product Type	5	0.894	0.828	0.805	0.895	0.828	0.805	0.897	0.830	0.808	0.829	0.754	0.728	0.687	0.628	0.608
	10	0.923	0.837	0.809	0.924	0.838	0.809	0.926	0.840	0.812	0.871	0.768	0.734	0.725	0.640	0.613
	20	0.945	0.843	0.811	0.946	0.843	0.811	0.947	0.845	0.813	0.904	0.776	0.736	0.761	0.649	0.616
All-Brands	5	0.350	0.294	0.276	0.362	0.303	0.283	0.359	0.301	0.282	0.286	0.221	0.200	0.287	0.253	0.241
	10	0.395	0.309	0.282	0.412	0.319	0.290	0.408	0.317	0.289	0.350	0.242	0.208	0.309	0.260	0.244
	20	0.439	0.320	0.285	0.459	0.331	0.293	0.453	0.329	0.292	0.408	0.256	0.212	0.330	0.265	0.246
Top-10k Brands	5	0.587	0.486	0.452	0.590	0.491	0.458	0.593	0.490	0.456	0.553	0.444	0.407	0.394	0.344	0.327
	10	0.667	0.512	0.463	0.669	0.517	0.469	0.674	0.517	0.467	0.645	0.473	0.420	0.433	0.357	0.333
	20	0.739	0.530	0.468	0.742	0.535	0.474	0.744	0.535	0.472	0.724	0.493	0.425	0.470	0.366	0.335

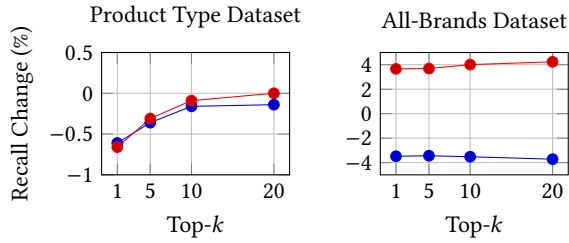

Figure 3: Impact of session length on Recall relative to the MUSCLE baseline (10 events): 5 events (red), 20 events (blue).

Table 2 shows the computation overhead for MUSCLE compared to the Query-Only model. By incorporating multiple events from the session in addition to the current query, MUSCLE increases the amount of data processed per sample. This results in a per-sample forward pass time that is roughly an order of magnitude higher than Query-Only. Compared to Session-Only models, the additional cost of integrating the current query alongside the session history adds a modest increase. In practice, this highlights an important consideration: while session-aware models can improve prediction quality, the trade-off in computation must be evaluated. Applications with strict latency requirements may rely on Query-Only or shorter sessions, while richer context can be selectively incorporated.

4.2 Qualitative Analysis

We examined the queries in which MUSCLE outperformed the baseline. The most frequent categories where MUSCLE showed improvements for product type prediction included: 1) brand-only queries, 2) broad terms, and 3) codes or cryptic names unfamiliar to the embedding model, such as e-commerce numeric identifiers. For brand prediction, the model relied on session-level preferences instead of popularity. For example, for broad queries like “*shoes*”,

Table 2: Inference computation overhead per 1 query, relative to the Query-Only baseline (1×), averaged over three runs.

Dataset	Query-Only	MUSCLE	Session-Only
Brands	1×	21.6×	17.2×
PTS	1×	23.3×	18.5×

Table 3: Session examples with ambiguity and intent shifts. Brand names are anonymized with <brand_x>, <brand_y>.

Session 1			Session 2		
Action	Query	PT	Action	Query	PT
click	<brand_x> yoga	BRA	search	<brand_y> conditioner	–
search	<brand_x> yoga	–	click	<brand_y> conditioner	HAIR COND.
click	<brand_x> yoga	BRA	search	<brand_y> honey	–
external	–	–	search	<brand_y> honey cond.	–
external	–	PANTS	search	shampoo <brand_y>	–
click	<brand_x> yoga	BRA	click	shampoo <brand_y>	SHAMPOO
click	<brand_x> yoga	BRA	search	makeup cotton pad	–
click	<brand_x> yoga	BRA	purchase	makeup cotton pad	COTTON PAD
click	<brand_x> yoga	BRA			
search	<brand_x> yoga	–			
purchase	<brand_x> yoga	BRA			

where e-commerce systems typically surface popular brands, MUSCLE recommended brands aligned with the user’s session context.

Consider the examples in Table 3 for product type prediction. For Session 1, the corresponding top-1 predictions for each method are: Query-Only: SHIRT; MUSCLE: BRA. In this case, in the query, we have a broad term and a brand that do not hint at a specific Product Type, illustrating a clear ambiguity case where MUSCLE helps. For session 2, the corresponding predictions for each method are: Session-Only: SHAMPOO; MUSCLE: COTTON PAD. This shows us a scenario of where MUSCLE helps against the session-only model since the last query indicates a shift in intent. Identifying future cases where MUSCLE outperforms competing models is crucial, as it allows us to trigger MUSCLE only for queries of this type, avoiding any computation overhead on queries that already have a strong signal for product type or brand.

5 Conclusion

We presented MUSCLE, a session-aware query classification model that integrates query semantics with session context, enabling the model to capture both immediate intent and evolving user preferences within a shopping session. Experimental results demonstrate that this design consistently outperforms state-of-the-art baselines on brand and product type prediction tasks, offering a practical approach for context-aware query understanding in e-commerce systems. Future work involves automatically identifying queries where MUSCLE outperforms session-less query understanding models, enabling selective use to avoid latency increase for queries with clear product type or brand signals.

References

- [1] Xusong Chen, Dong Liu, Chenyi Lei, Rui Li, Zheng-Jun Zha, and Zhiwei Xiong. 2019. BERT4SessRec: Content-Based Video Relevance Prediction with Bidirectional Encoder Representations from Transformer. In *ACM Multimedia*. ACM, 2597–2601.
- [2] Gabriel de Souza Pereira Moreira, Sara Rabhi, Jeongmin Lee, Ronay Ak, and Even Oldridge. 2021. Transformers4Rec: Bridging the Gap between NLP and Sequential / Session-Based Recommendation. In *RecSys*. ACM, 143–153.
- [3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL-HLT (1)*. Association for Computational Linguistics, 4171–4186.
- [4] Jun Fang. 2021. Session-based Recommendation with Self-Attention Networks. *CoRR* abs/2102.01922 (2021).
- [5] Balázs Hidasi, Alexandros Karatzoglou, Linas Baltrunas, and Domonkos Tikk. 2016. Session-based Recommendations with Recurrent Neural Networks. In *ICLR (Poster)*.
- [6] Jiayi Hu, Jingtong Gao, Xiangyu Zhao, Yuehong Hu, Yuxuan Liang, Yiqi Wang, Ming He, Zitao Liu, and Hongzhi Yin. 2024. BiVRec: Bidirectional View-based Multimodal Sequential Recommendation. *CoRR* abs/2402.17334 (2024).
- [7] Akshay Jagatap, Nikki Gupta, Sachin Farfade, and Prakash Mandayam Comar. 2023. AttriBERT - Session-based Product Attribute Recommendation with BERT. In *SIGIR*. ACM, 3421–3425.
- [8] Wei Ji, Xiangyan Liu, An Zhang, Yinwei Wei, Yongxin Ni, and Xiang Wang. 2023. Online Distillation-enhanced Multi-modal Transformer for Sequential Recommendation. In *ACM Multimedia*. ACM, 955–965.
- [9] Wang-Cheng Kang and Julian J. McAuley. 2018. Self-Attentive Sequential Recommendation. In *ICDM*. IEEE Computer Society, 197–206.
- [10] T. Liu and S. Nag. 2025. Query-to-Brand Entity Linking for Short and Ambiguous E-Commerce Queries. In *arXiv preprint arXiv:2502.01234*. <https://arxiv.org/abs/2502.01234>
- [11] Yong Liu, Susen Yang, Chenyi Lei, Guoxin Wang, Haihong Tang, Juyong Zhang, Aixin Sun, and Chunyan Miao. 2021. Pre-training Graph Transformer with Multimodal Side Information for Recommendation. In *ACM Multimedia*. ACM, 2853–2861.
- [12] Michael Skinner and Surya Kallumadi. 2019. E-commerce Query Classification Using Product Taxonomy Mapping: A Transfer Learning Approach. In *eCOM@SIGIR (CEUR Workshop Proceedings, Vol. 2410)*. CEUR-WS.org.
- [13] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. 2019. BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer. In *CIKM*. ACM, 1441–1450.
- [14] Anna Tiginova, Thomas Ricatte, and Ghadir Eraisha. 2024. Transfer Learning for E-commerce Query Product Type Prediction. *CoRR* abs/2410.07121 (2024).
- [15] Jinpeng Wang, Ziyun Zeng, Yunxiao Wang, Yuting Wang, Xingyu Lu, Tianxiang Li, Jun Yuan, Rui Zhang, Hai-Tao Zheng, and Shu-Tao Xia. 2023. MISSRec: Pre-training and Transferring Multi-modal Interest-aware Sequence Representation for Recommendation. In *ACM Multimedia*. ACM, 6548–6557.
- [16] Shuhan Wang, Bin Shen, Xu Min, Yong He, Xiaolu Zhang, Liang Zhang, Jun Zhou, and Linjian Mo. 2024. Aligned Side Information Fusion Method for Sequential Recommendation. In *WWW (Companion Volume)*. ACM, 112–120.
- [17] Yueqi Xie, Peilin Zhou, and Sunghun Kim. 2022. Decoupled Side Information Fusion for Sequential Recommendation. In *SIGIR*. ACM, 1611–1621.
- [18] Shuai Zhang, Yi Tay, Lina Yao, and Aixin Sun. 2018. Next Item Recommendation with Self-Attention. *CoRR* abs/1808.06414 (2018).
- [19] Jiashu Zhao, Hongshen Chen, and Dawei Yin. 2019. A Dynamic Product-aware Learning Model for E-commerce Query Intent Understanding. In *CIKM*. ACM, 1843–1852.